

Practical 14

Shell script practicals

1. Create a shell script that checks if a file exists in the current directory.

```
student@student-virtual-machine:~/Downloads$ nano file_exists.sh
student@student-virtual-machine:~/Downloads$ chmod +x file_exists.sh
student@student-virtual-machine:~/Downloads$ ./file_exists.sh
File not found: example.txt
student@student-virtual-machine:~/Downloads$ nano file_exists.sh
student@student-virtual-machine:~/Downloads$ ./file_exists.sh
File exists: global_eCommerce_customer_analytics_report.txt
```

```
#!/bin/bash

file="global_eCommerce_customer_analytics_report.txt"

if [ -e "$file" ]; then
    echo "File exists: $file"
else
    echo "File not found: $file"
fi
```

2. How do you use the for loop to iterate through a list of values?

```
student@student-virtual-machine:~/Downloads$ nano example.sh
student@student-virtual-machine:~/Downloads$ chmod +x example.sh
student@student-virtual-machine:~/Downloads$ ./example.sh
Current fruit: apple
Current fruit: banana
Current fruit: cherry
Current fruit: date
```

```
#!/bin/bash

fruits=("apple" "banana" "cherry" "date")

for fruit in "${fruits[@]"; do
    echo "Current fruit: $fruit"
done
```

3. Create a script that searches for a specific word in a file and counts its occurrences.

```
student@student-virtual-machine:~/Downloads$ nano sample.txt
student@student-virtual-machine:~/Downloads$ nano word_count.sh
student@student-virtual-machine:~/Downloads$ bash word_count.sh
Enter the word to search for:
banana
Enter the filename:
sample.txt
The word 'banana' appears 2 times in 'sample.txt'.
```

```

nano nano 0.2
#!/bin/bash

echo "Enter the word to search for:"
read target_word

echo "Enter the filename:"
read filename

count=$(grep -o -w "$target_word" "$filename" | wc -l)

echo "The word '$target_word' appears $count times in '$filename'."

```

apple banana apple mango apple
banana apple orange

4. Write a script to calculate the sum of Elements in an Array.

```

student@student-virtual-machine:~/Downloads$ nano array_sum.sh
student@student-virtual-machine:~/Downloads$ bash array_sum.sh
The sum of elements in the array is: 201

```

```

nano nano 0.2
#!/bin/bash

array=(1 65 22 19 94)

sum=0

for num in "${array[@]}; do
    sum=$((sum + num))
done

echo "The sum of elements in the array is: $sum"

```

5. Performing Mathematical Operations Without Storing

```

student@student-virtual-machine:~/Downloads$ nano math_operations.sh
student@student-virtual-machine:~/Downloads$ bash math_operations.sh
Enter a number: 45
Enter a smaller number: 4
Addition: 49
Subtraction: 41
Multiplication: 180
Division: 11

```

```
GNU nano 0.2
#!/bin/bash

read -p "Enter a number: " num1
read -p "Enter a smaller number: " num2

echo "Addition: $((num1 + num2))"
echo "Subtraction: $((num1 - num2))"
echo "Multiplication: $((num1 * num2))"
echo "Division: $((num1 / num2))"
```

6. Check If a File is Writable

```
student@student-virtual-machine:~/Downloads$ nano check_writable.sh
student@student-virtual-machine:~/Downloads$ bash check_writable.sh
Enter a File Name: global_eCommerce_customer_analytics_report.txt
The File global_eCommerce_customer_analytics_report.txt is writable.
```

```
GNU nano 0.2
#!/bin/bash

read -p "Enter a File Name: " fname

if [ -w "$fname" ]; then
    echo "The File $fname is writable."
else
    echo "The File $fname is not writable."
fi
```

7. Check If a File Exists or Not

```
student@student-virtual-machine:~/Downloads$ nano check_file.sh
student@student-virtual-machine:~/Downloads$ bash check_file.sh
Enter a File Name: global_eCommerce_customer_analytics_report.txt
The File global_eCommerce_customer_analytics_report.txt exists.
student@student-virtual-machine:~/Downloads$ bash check_file.sh
Enter a File Name: hello
The File hello does not exist!
```

```
#!/bin/bash

read -p "Enter a File Name: " fname

if [ ! -f "$fname" ]; then
    echo "The File $fname does not exist!"
    exit 1
fi

echo "The File $fname exists."
```

8. Create a New File and Write Text Inside

```
student@student-virtual-machine:~/Downloads$ nano create_file.sh
student@student-virtual-machine:~/Downloads$ bash create_file.sh
Enter the file name: hello
Enter text to write:
wonderful color of life ,
-----
The File hello is created!
```

```
#!/bin/bash

read -p "Enter the file name: " file

echo "Enter text to write:"
read text

echo "$text" > "$file"

echo "-----"
echo "The File $file is created!"
```

9. Check the Permissions of a File

```
#!/bin/bash

read -p "Enter the file name: " file

if [ -f "$file" ]; then

    if [ -r "$file" ]; then
        echo "Readable"
    fi

    if [ -w "$file" ]; then
        echo "Writable"
    fi

    if [ -x "$file" ]; then
        echo "Executable"
    fi

else
    echo "Error! The file $file does not exist."
fi
```

```
student@student-virtual-machine:~/Downloads$ nano check_permissions.sh
student@student-virtual-machine:~/Downloads$
student@student-virtual-machine:~/Downloads$ bash check_permissions.sh
Enter the file name: global_eCommerce_customer_analytics_report.txt
Readable
Writable
```

10. Sets the Permissions of a Directory for the Owner

```
#!/bin/bash

read -p "Enter the directory name: " dir

if [ -d "$dir" ]; then
    chmod u+rx "$dir"
    echo "Directory permissions have been updated!"
else
    echo "Error! The directory $dir does not exist."
fi
```

```
student@student-virtual-machine:~/Downloads$ nano directory_permission.sh
student@student-virtual-machine:~/Downloads$ mkdir mydirectory
student@student-virtual-machine:~/Downloads$ bash directory_permission.sh
Enter the directory name: mydirectory
Directory permissions have been updated!
student@student-virtual-machine:~/Downloads$ ls -ld mydirectory
drwxrwxr-x 2 student student 4096 Aug 24 18:41 mydirectory
```

11. Check If a Directory Exists or Not

```
#!/bin/bash

read -p "Enter a Directory Name: " dir

if [ ! -d "$dir" ]; then
    echo "The directory $dir does not exist!"
    exit 1
fi

echo "The directory $dir exists."
```

```
student@student-virtual-machine:~/Downloads$ nano check_directory.sh
student@student-virtual-machine:~/Downloads$ bash check_directory.sh
Enter a Directory Name: mydirectory
The directory mydirectory exists.
student@student-virtual-machine:~/Downloads$ bash check_directory.sh
Enter a Directory Name: nanopop
The directory nanopop does not exist!
```

12. Print Numbers From 5 to 1

```
#!/bin/bash

n=5

until [ "$n" -eq 0 ]
do
    echo "$n"
    n=$((n - 1))
done
```

```
student@student-virtual-machine:~/Downloads$ nano reverse_numbers.sh
student@student-virtual-machine:~/Downloads$
student@student-virtual-machine:~/Downloads$ bash reverse_numbers.sh
5
4
3
2
1
```

13. Calculate the Sum of Digits of a Given Number

```
#!/bin/bash

read -p "Enter a number: " num

sum=0

while [ "$num" -gt 0 ]
do
    dig=$((num % 10))
    sum=$((sum + dig))
    num=$((num / 10))
done

echo "The sum of digits of the given number: $sum"
```

```
student@student-virtual-machine:~/Downloads$ nano sum_digits.sh
student@student-virtual-machine:~/Downloads$ bash sum_digits.sh
Enter a number: 45632
The sum of digits of the given number: 20
```

14. Loop Through Array Elements

```
#!/bin/bash

arr=("mango" "grape" "apple" "cherry" "orange")

for item in "${arr[@]}"; do
    echo "$item"
done
```

```
student@student-virtual-machine:~/Downloads$ nano loop_array.sh
student@student-virtual-machine:~/Downloads$ bash loop_array.sh
mango
grape
apple
cherry
orange
```